

10(a)(i)	introduction of tracer (into the body)	M1
	containing a β^+ emitter	A1
10(a)(ii)	positron interacts with electron	B1
	(pair) annihilation occurs	B1
	mass of particles converted into gamma photons	B1
10(b)	(annihilation of electron and positron) produces two photons	B1
	$E = (\Delta)mc^2$	B1
	$E = hf$ and $f = c / \lambda$ or $E = hc / \lambda$	B1
	$\lambda = \{[2\times] 6.63 \times 10^{-34} \times 3.00 \times 10^8\} / \{[2\times] 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2\}$ $= 2.4(3) \times 10^{-12} \text{ m or } 2.4(3) \text{ pm (full substitution and answer with unit needed)}$	B1